PROJECT DOCUMENTATION

LumenIndex
Rural Development Index for Latin America

A unified reference to the project's purpose, data foundation, analytical methodology, applications, models, engineering controls, and delivered assets.

BUILT FOR

Living Stones Foundation
Applied Data & Digital Innovation Lab (LATAM)

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PROJECT GEOGRAPHY

Latin America
17-country combined dataset

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Project purpose

LumenIndex transforms dispersed public development data into a consistent analytical experience for comparing economic, social, infrastructure, labor, poverty, and agricultural-research conditions across Latin America.

DOCUMENT GUIDE

One reference for the complete project

This document describes what LumenIndex delivers and how its parts work together. It is written for program, analytics, data, and engineering readers and focuses on the finished project package rather than review history or transfer activities.

> **Executive summary**
>
> The project combines public datasets, Neon PostgreSQL, a Streamlit dashboard, a guarded natural-language data agent, an XGBoost poverty workflow, a deterministic development-tier rule, data-integrity SQL, and reproducible setup documentation.

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Primary users

Program teams, researchers, analysts, and technical teams working with rural-development evidence in Latin America.

Primary outputs

Interactive exploration, filtered data exports, natural-language answers, poverty-model evaluation, and transparent development tiers.

01 / PROJECT SCOPE

LumenIndex turns fragmented indicators into a connected evidence product

Rural development is multidimensional. Economic output alone does not describe access, opportunity, resilience, or research capacity. LumenIndex organizes complementary indicators into one system so a reader can move from a regional overview to country-level patterns and source values without switching tools.

17
countries
represented in the combined analytical dataset.

11
available indicators
spanning labor, research, economy, access, health, poverty, and population.

3
public-data families
World Bank WDI, female labor force data, and IFPRI ASTI.

Project objectives

- Consolidate development indicators into a structured cloud data layer.
- Provide an accessible visual interface for filtering, comparing, and exporting observed data.
- Offer plain-language access to data through a natural-language AI agent.
- Create a transparent composite score and consistent High, Medium, and Low development tiers.
- Support analytical experimentation with a reproducible poverty-prediction workflow.
- Embed security, integrity, and provenance controls in the implementation.

Project outcomes

For program teams
A concise view of country patterns and changes over time, with source gaps kept visible.

For analysts
Structured tables, filtered exports, documented score logic, and model scripts that can be inspected and rerun.

For technical teams
Environment-based configuration, database safeguards, reproducible dependencies, and modular application code.

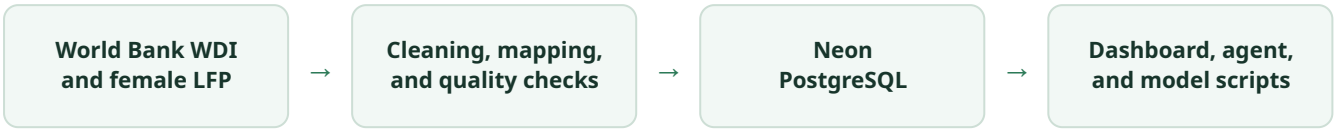
For stakeholders
Two complementary access modes: interactive visual analysis and plain-language questions.

The package covers data, applications, analytics, and documentation

Deliverable	Purpose	What it provides
Streamlit dashboard	Interactive analysis	Country, year, and tier filters; indicators; trends; rankings; tables; observed-data exports.
LumenIndex AI Agent	Natural-language access	Plain-English questions, guarded data retrieval, evidence table, answer labeling, and visible generated query.
Neon PostgreSQL data layer	Central storage	Five documented tables for WDI, female labor, ASTI, poverty, and combined analysis.
Composite index	Comparable score	Normalized weighted score from 0 to 100 and deterministic High, Medium, and Low tiers.
Model 1	Poverty estimation	XGBoost regression, leakage-aware preprocessing, time-based evaluation, grouped cross-validation, and feature importance.
Model 2	Tier assignment	An auditable script that reproduces the fixed score thresholds.
Data and integrity tooling	Reproducibility	Multi-country WDI retrieval, percentage checks, uniqueness controls, and read-only database-role SQL.
Supporting assets	Project communication	Excel workbooks, dashboard preview, saved model file, project presentation, README, and dependency manifest.

> **Authoritative repository**
>
> https://github.com/Applied-Data-Digital-Innovation-Lab/lumenindex-latam

Public data flows through a PostgreSQL-centered analytical stack



Source families

- **World Development Indicators:** economy, infrastructure access, health, poverty, and population.
- **Female labor force participation:** longitudinal labor participation across Latin America.
- **IFPRI ASTI:** agricultural research spending, researcher capacity, and PhD composition.

Application boundaries

- The dashboard reads the combined table and computes score outputs for visualization.
- The agent reads documented tables through validated, read-only query execution.
- Model scripts connect to the same combined table using environment-based configuration.

Documented cloud tables

Table	Rows	Analytical role
`chile_development_indicators`	298	Chile WDI indicators
`female_labor_force_latam`	648	Female labor force participation
`asti_agricultural_research`	2,095	Agricultural research indicators
`lumenindex_combined`	527	Composite-score and model dataset
`latam_poverty`	194	Poverty headcount series

> **Design principle**
>
> Observed values remain available for charts, tables, and downloads. Values imputed for score calculation are stored in separate working columns so missingness is not silently rewritten as observation.

The score is transparent, weighted, and reproducible

The application begins with 11 available indicators. Median imputation is applied only to score-working columns. Each indicator is min-max normalized to a 0-100 scale. For measures where a lower value is favorable, the normalized direction is reversed.

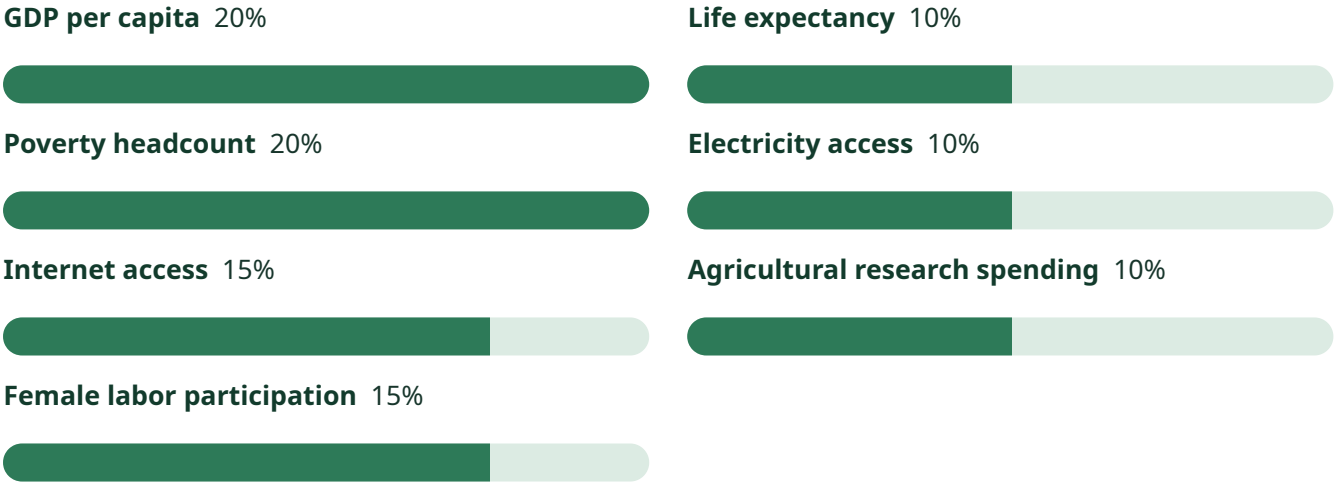
Higher is favorable

Female labor force participation, agricultural research spending, full-time-equivalent researchers, researchers with PhDs, GDP per capita, electricity access, internet access, and life expectancy.

Lower is favorable

Unemployment rate, poverty headcount, and rural population percentage.

Weighted score inputs



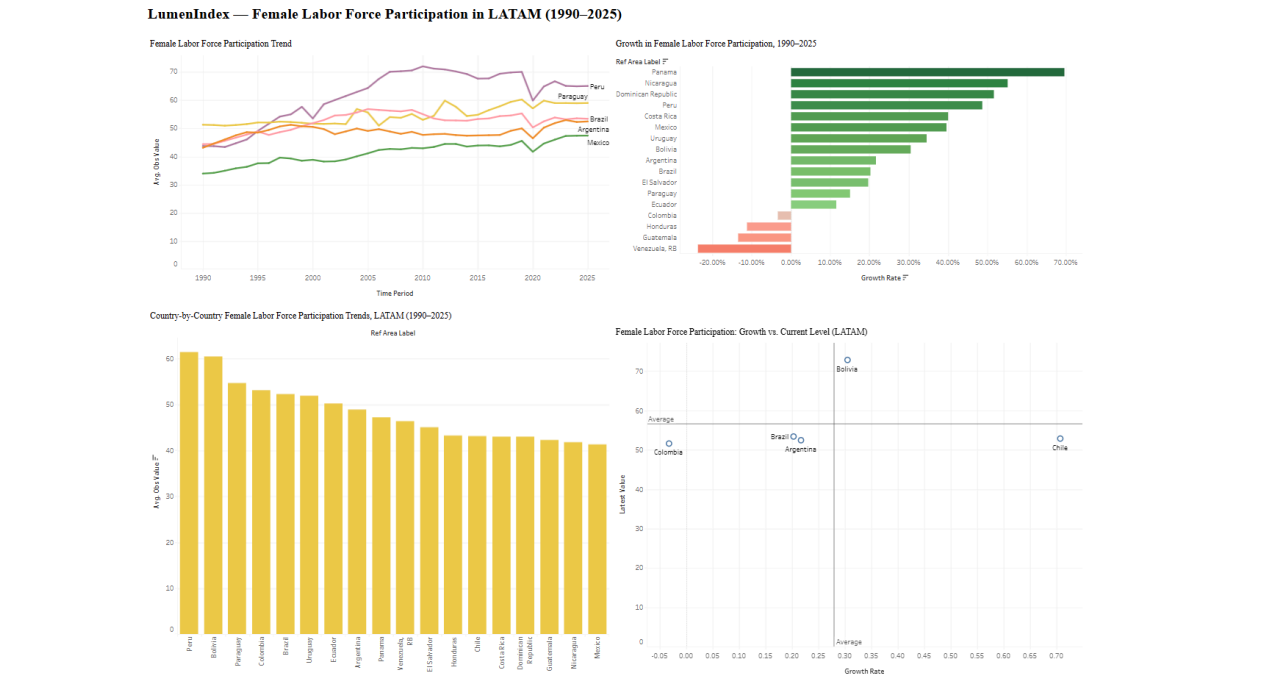
Development tiers

High	Composite score of 60 or above.
Medium	Composite score from 35 to below 60.
Low	Composite score below 35.

> **Interpretation**
>
> The score is a comparative analytical construct based on the selected indicators, normalization range, weights, and data coverage. It is not an official socioeconomic classification.

The dashboard moves from regional overview to source-level detail

The Streamlit interface is designed for non-technical exploration. Users can filter the analytical view by countries, year range, and development tier, then move between key indicators, rankings, trends, individual measures, poverty views, and underlying data.



LumenIndex dashboard preview. The interface combines summary metrics, interactive filters, charts, and data tables.

Explore

Country, year, and tier controls update the visible analysis.

Compare

Latest-country views support rankings and side-by-side interpretation.

Export

Downloadable outputs preserve observed gaps instead of replacing them with score imputations.

Published experiences

LumenIndex dashboard:
https://lumenindex-latam01.streamlit.app/

LATAM Female Labor Force dashboard:
https://public.tableau.com/app/profile/deepeka.gurunathan/viz/LATAMLaborForceDashboard/Dashboard1?publish=yes

Plain-language questions are answered through guarded data access

The LumenIndex AI Agent lets stakeholders ask questions without writing SQL. The application generates a database query, validates it, runs it inside a controlled read-only transaction, and turns returned rows into a plain-English answer. The result table and generated query remain available for inspection.

User question → **Query generation** → **Safety validation** → **Data result and plain-English answer**

Defense in depth

Control	Implemented behavior
Access gate	Application access is protected by `AGENT_PASSCODE`.
Statement type	Only a single query beginning with `SELECT` or `WITH` is accepted.
Forbidden operations	Write, DDL, privilege, maintenance, procedure, and multi-statement patterns are rejected.
Database transaction	Execution begins with `SET TRANSACTION READ ONLY`.
Resource limits	Database execution has a 5-second statement timeout and a 200-row outer cap.
Transparency	The interface distinguishes database-backed answers and exposes the generated query.
Least privilege	The integrity SQL includes a read-only role pattern with CONNECT, schema USAGE, and SELECT grants.

Example use

Compare female labor force participation across countries or examine change within a country over time.

Conversation context

The application maintains the most recent 10 exchanges to support follow-up questions.

Published agent:
https://lumenindexagent-latam.streamlit.app/

Two workflows serve different analytical purposes

Model 1: poverty predictor

An XGBoost regression workflow estimates poverty headcount from 10 development measures plus year.

- Loads non-null target rows from `lumenindex_combined`.
- Uses median imputation inside a scikit-learn Pipeline.
- Uses a time split: train through 2015 and test after 2015 when the year range permits.
- Falls back to a country-grouped split when a time split is not possible.
- Runs grouped cross-validation by country.
- Reports R^2, root mean squared error, mean absolute error, and feature importance.
- Supports estimates for rows where poverty headcount is missing.

Model 2: development tier rule

A deterministic workflow recomputes the composite score and assigns the same tiers used by the dashboard.

- Reads the 11 index indicators from the combined table.
- Applies median imputation to working data.
- Normalizes favorable and unfavorable measures in the correct direction.
- Applies the documented seven-indicator weights.
- Assigns High, Medium, or Low from fixed score thresholds.
- Prints the latest country result and tier counts for inspection.

> **Why the distinction matters**
>
> Model 1 is predictive and must be evaluated on held-out data. Model 2 is not a learned classifier; it is an auditable implementation of the project's policy rule, so classifier-accuracy claims do not apply.

Evaluation integrity

Keeping imputation inside the training pipeline prevents information from the evaluation set from shaping the preprocessing statistics. Separating later years for testing and grouping cross-validation by country reduces leakage between closely related observations.

Output practice

Model 1 performance values are produced by the script from the connected dataset rather than hard-coded in this document. This keeps reported results tied to the exact data snapshot and split used for a run.

The repository is structured for reproducible operation

Core files

`app.py`	Streamlit dashboard
`agent.py`	Natural-language data agent
`Model1_poverty_predictor.py`	Poverty model workflow
`model2_tier_classifier.py`	Deterministic tier rule
`requirements.txt`	Python dependencies
`model1_poverty_predictor.json`	Saved XGBoost model asset

Data and controls

`scripts/fetch_wdi_all_latam.py`	Seven-indicator, 17-country WDI retrieval
`sql/migrations/001_data_integrity.sql`	Percentage check, ASTI uniqueness, read-only role pattern
`chile_lumenindex.xlsx`	Chile Excel analysis
`WB_WDI_SL_TLF_CACT_FE_ZS.xlsx`	Female labor force dataset
`lumenindex_dashboard.png`	Dashboard preview
`README.md`	Project reference

Runtime configuration

```
pip install -r requirements.txt
python -m streamlit run app.py
python -m streamlit run agent.py
```

`DATABASE_URL`, `LSF_TOKEN`, and `AGENT_PASSCODE` are supplied through environment or Streamlit secret storage; credential values are not embedded in the codebase.

Interpretation boundaries

- Several World Bank indicators in the combined dataset have materially better coverage for Chile than for other countries. Cross-country rankings are best treated as analytical demonstrations where coverage is uneven.
- ASTI source data includes mixed-unit duplicate series and an upstream field-mapping risk. The repository includes range and uniqueness controls to make these conditions visible.
- The composite score depends on the current indicator set, normalization range, weights, imputation method, and available observations.
- The tier workflow is deterministic; it does not predict an independently labeled outcome.

Project and source references

Repository: https://github.com/Applied-Data-Digital-Innovation-Lab/lumenindex-latam
World Bank: https://data.worldbank.org
IFPRI ASTI: https://asti.cgiar.org
Poverty indicator: https://data.worldbank.org/indicator/SI.POV.NAHC